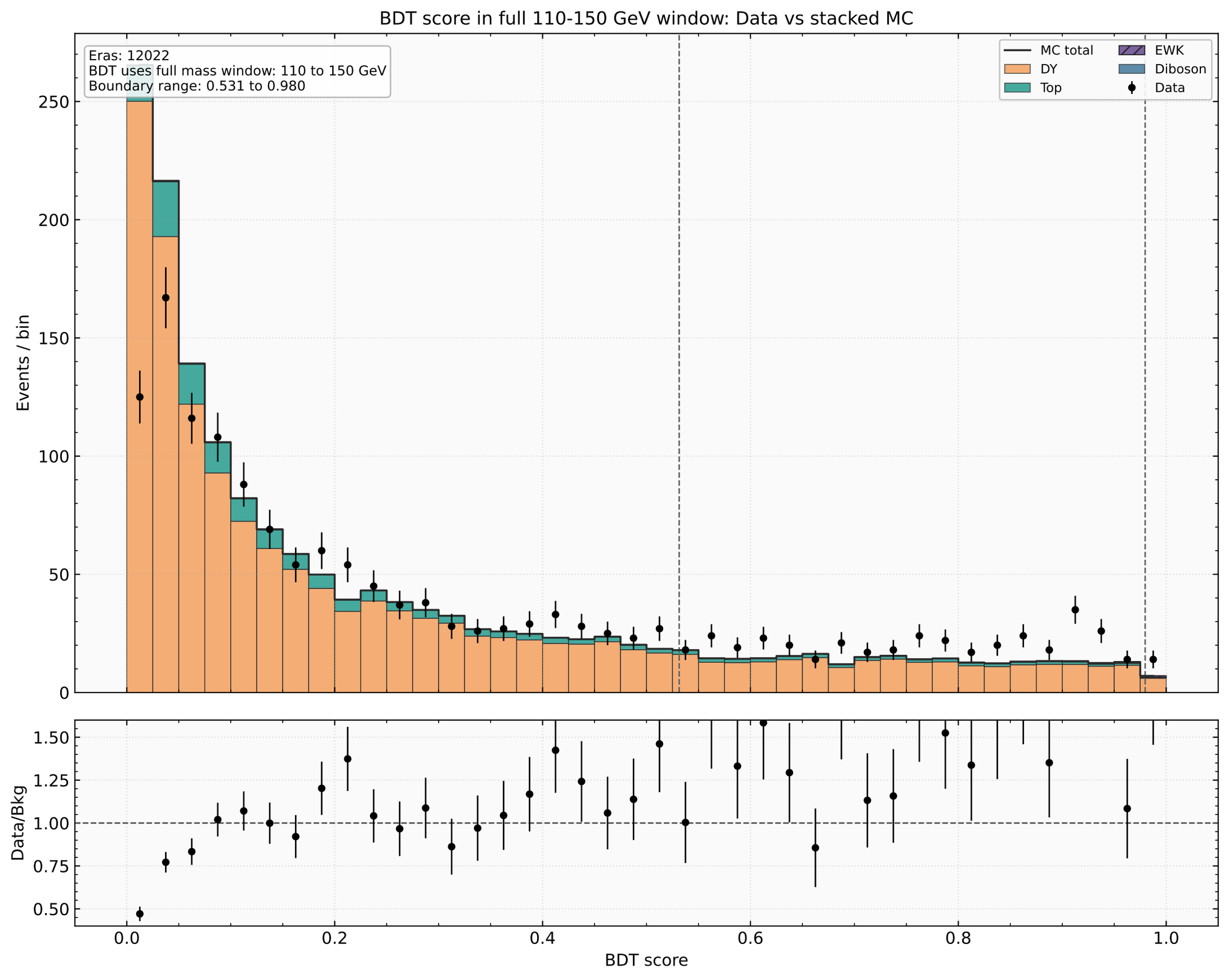
Run-3 BDT Sideband Validation Report

Outputs directory: /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Eras used: 12022
Extra cut: (none)
Blinding window: 115 < m_mumu < 135 GeV
Sidebands: (110, 115) U (135, 150) GeV
Optimized category edges: 0.0000, 0.5314, 0.7869, 0.8982, 0.9522, 0.9797, 1.0000

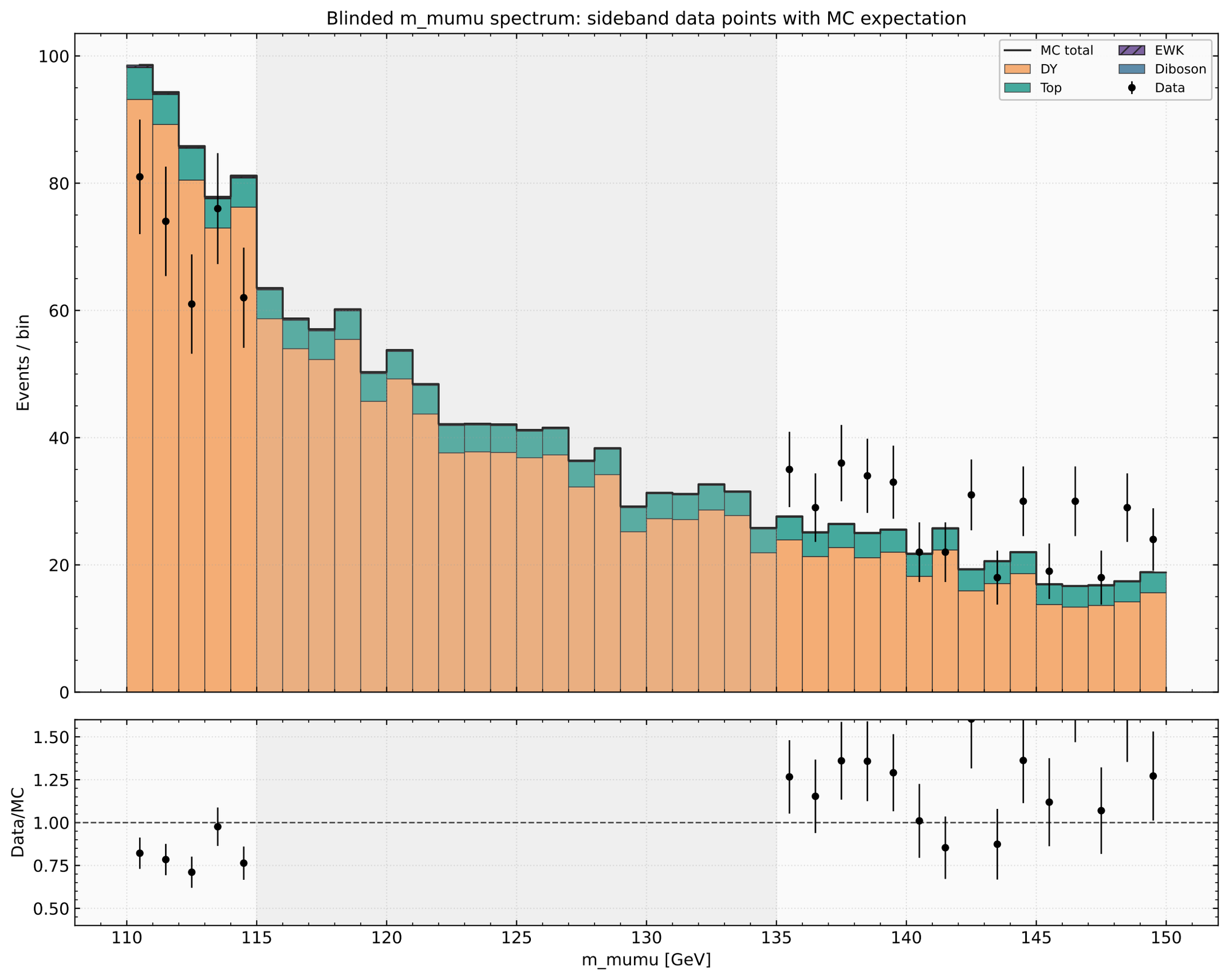

BDT score in full 110-150 GeV window: Data vs stacked MC
Eras: 12022
BDT uses full mass window: 110 to 150 GeV
Boundary range: 0.531 to 0.980
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data/Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

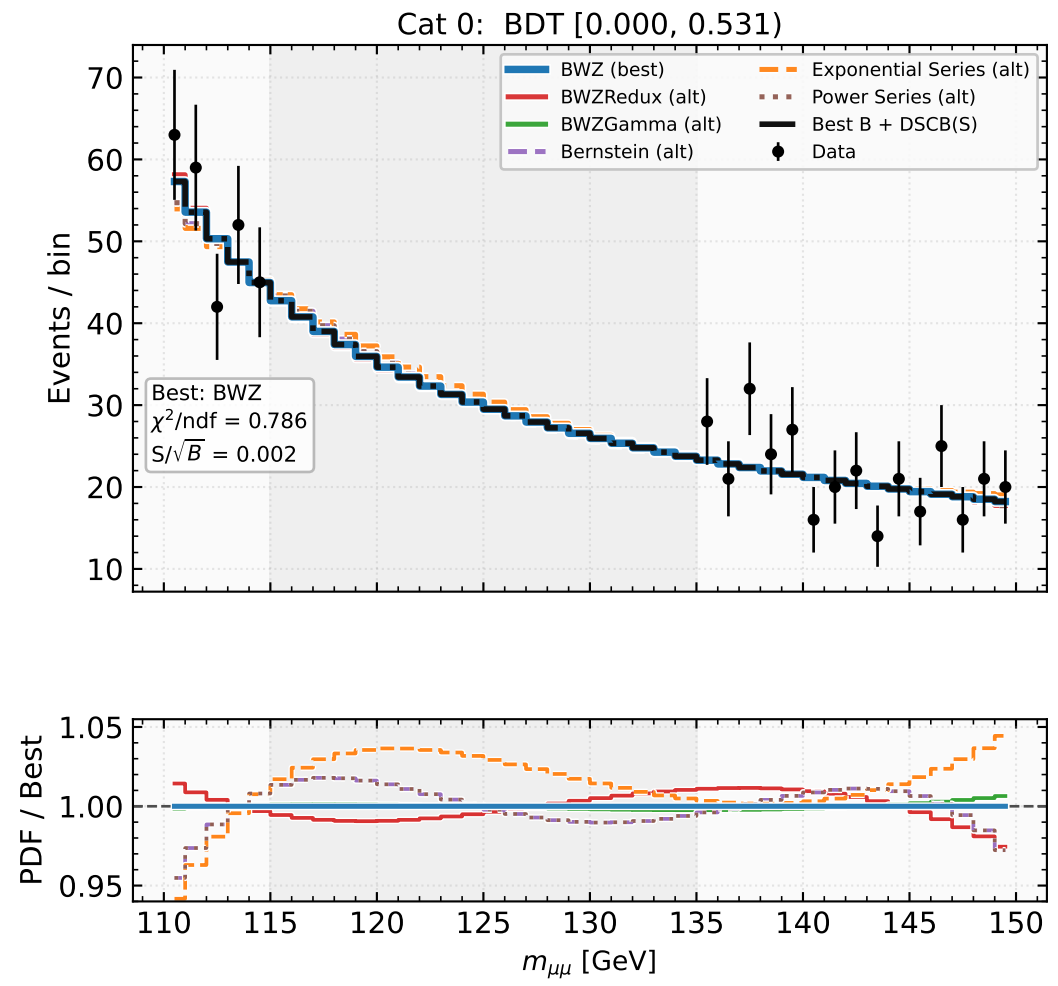
BDT bin	Total signal	Bkg. ± DeltaB	Data	S/(S+B) (%)	S/sqrt(B)
[0.000, 0.025)	0.001	265.49 ± 6.86	125	0.00	0.000
[0.025, 0.050)	0.002	216.45 ± 6.17	167	0.00	0.000
[0.050, 0.075)	0.002	139.18 ± 4.94	116	0.00	0.000
[0.075, 0.100)	0.002	105.89 ± 4.36	108	0.00	0.000
[0.100, 0.125)	0.002	82.21 ± 3.84	88	0.00	0.000
[0.125, 0.150)	0.002	69.05 ± 3.53	69	0.00	0.000
[0.150, 0.175)	0.002	58.63 ± 3.27	54	0.00	0.000
[0.175, 0.200)	0.002	49.90 ± 3.02	60	0.00	0.000
[0.200, 0.225)	0.002	39.31 ± 2.64	54	0.01	0.000
[0.225, 0.250)	0.002	43.19 ± 2.81	45	0.01	0.000
[0.250, 0.275)	0.002	38.27 ± 2.70	37	0.01	0.000
[0.275, 0.300)	0.002	34.94 ± 2.56	38	0.01	0.000
[0.300, 0.325)	0.002	32.47 ± 2.48	28	0.01	0.000
[0.325, 0.350)	0.003	26.79 ± 2.15	26	0.01	0.000
[0.350, 0.375)	0.002	25.85 ± 2.20	27	0.01	0.000
[0.375, 0.400)	0.003	24.82 ± 2.16	29	0.01	0.001
[0.400, 0.425)	0.003	23.17 ± 2.12	33	0.01	0.001
[0.425, 0.450)	0.003	22.54 ± 2.08	28	0.01	0.001
[0.450, 0.475)	0.003	23.62 ± 2.12	25	0.01	0.001
[0.475, 0.500)	0.003	20.21 ± 1.92	23	0.02	0.001

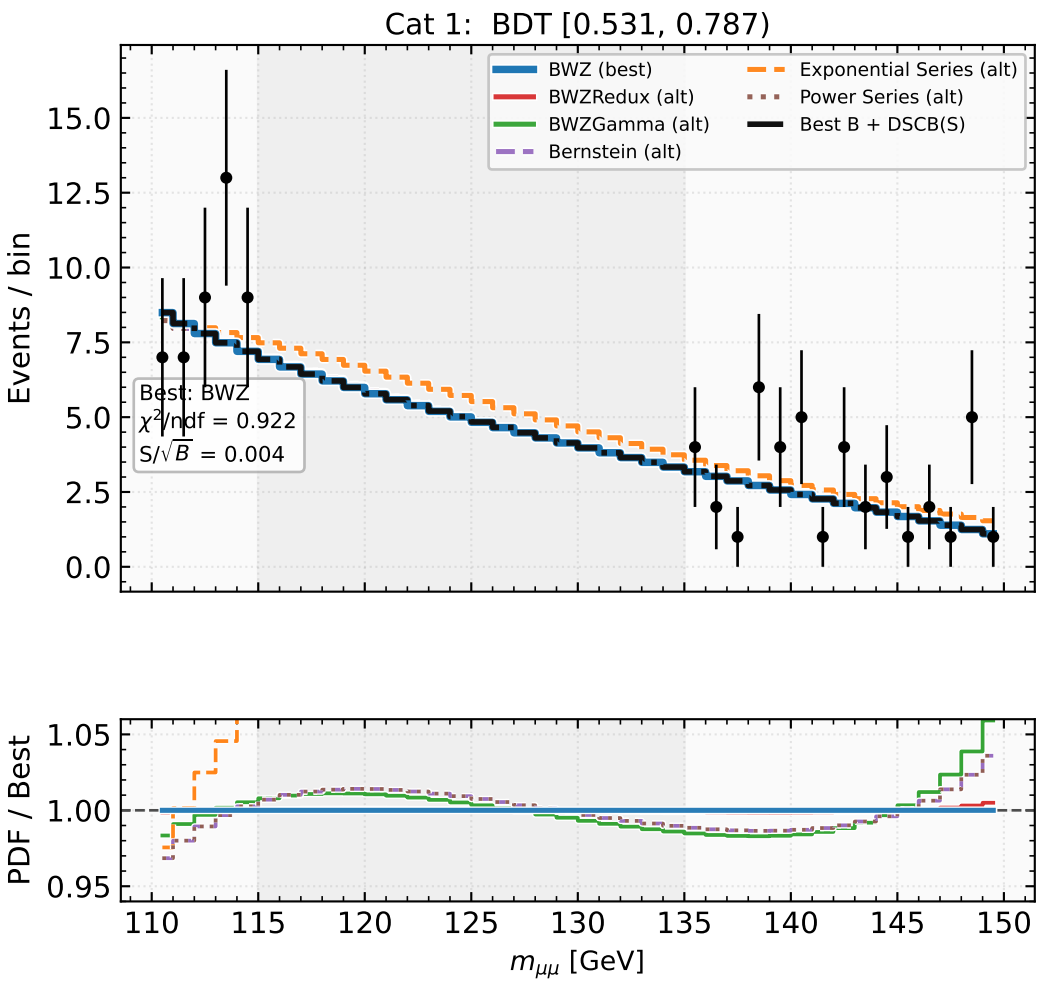
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

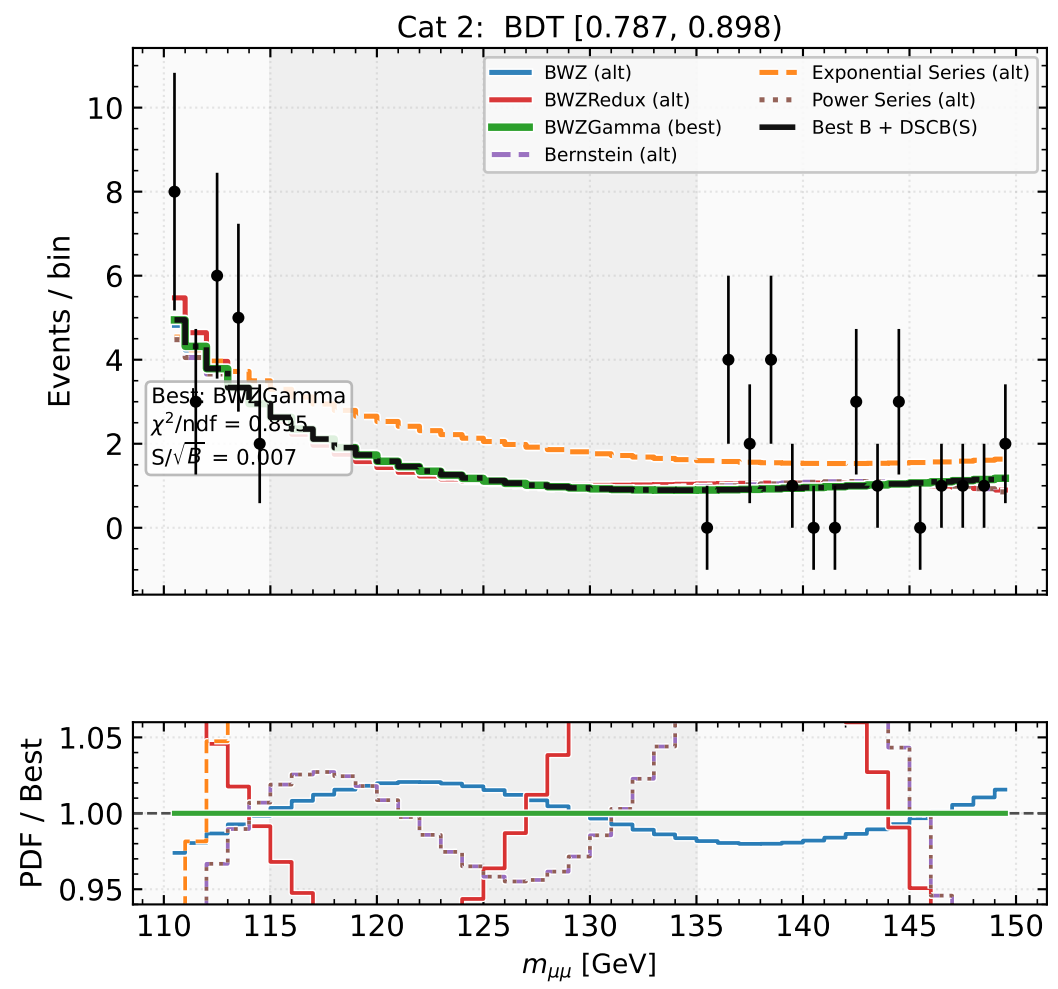
BDT bin	Total signal	Bkg. ± DeltaB	Data	S/(S+B) (%)	S/sqrt(B)
[0.500, 0.525)	0.003	18.48 ± 1.83	27	0.02	0.001
[0.525, 0.550)	0.004	17.94 ± 1.78	18	0.02	0.001
[0.550, 0.575)	0.004	14.51 ± 1.66	24	0.02	0.001
[0.575, 0.600)	0.004	14.27 ± 1.56	19	0.03	0.001
[0.600, 0.625)	0.004	14.52 ± 1.62	23	0.03	0.001
[0.625, 0.650)	0.005	15.46 ± 1.65	20	0.03	0.001
[0.650, 0.675)	0.005	16.36 ± 1.74	14	0.03	0.001
[0.675, 0.700)	0.005	11.97 ± 1.48	21	0.04	0.002
[0.700, 0.725)	0.006	15.02 ± 1.70	17	0.04	0.002
[0.725, 0.750)	0.007	15.55 ± 1.70	18	0.04	0.002
[0.750, 0.775)	0.008	14.08 ± 1.61	24	0.05	0.002
[0.775, 0.800)	0.008	14.43 ± 1.68	22	0.06	0.002
[0.800, 0.825)	0.010	12.71 ± 1.51	17	0.08	0.003
[0.825, 0.850)	0.011	12.36 ± 1.48	20	0.09	0.003
[0.850, 0.875)	0.014	13.09 ± 1.56	24	0.10	0.004
[0.875, 0.900)	0.017	13.32 ± 1.53	18	0.13	0.005
[0.900, 0.925)	0.021	13.28 ± 1.63	35	0.16	0.006
[0.925, 0.950)	0.029	12.49 ± 1.51	26	0.23	0.008
[0.950, 0.975)	0.046	12.91 ± 1.53	14	0.35	0.013
[0.975, 1.000)	0.067	7.05 ± 1.14	14	0.94	0.025

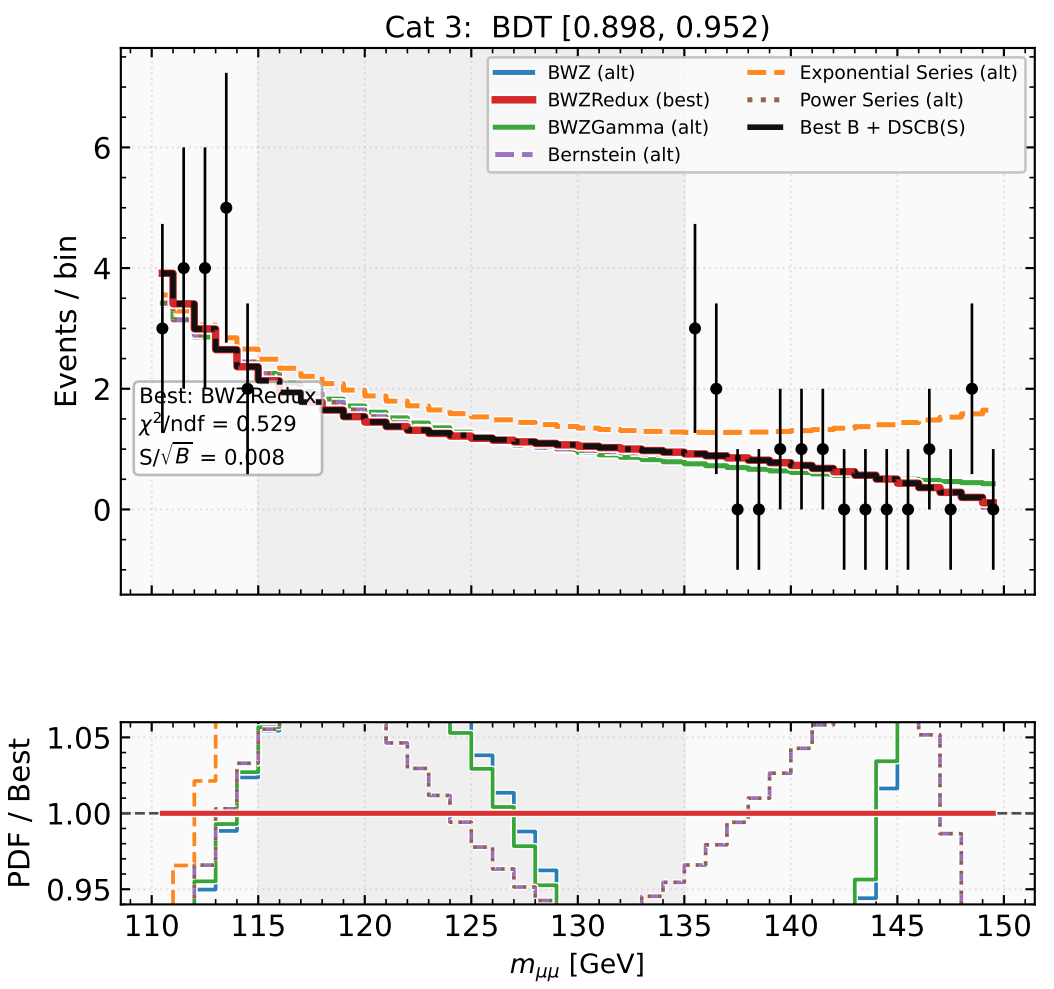

Blinded m_mumu spectrum: sideband data points with MC expectation
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
m_mumu [GeV]
0
20
40
60
80
100
0.50
0.75
1.00
1.25
1.50
110
115
120
125
130
135
140
145
150

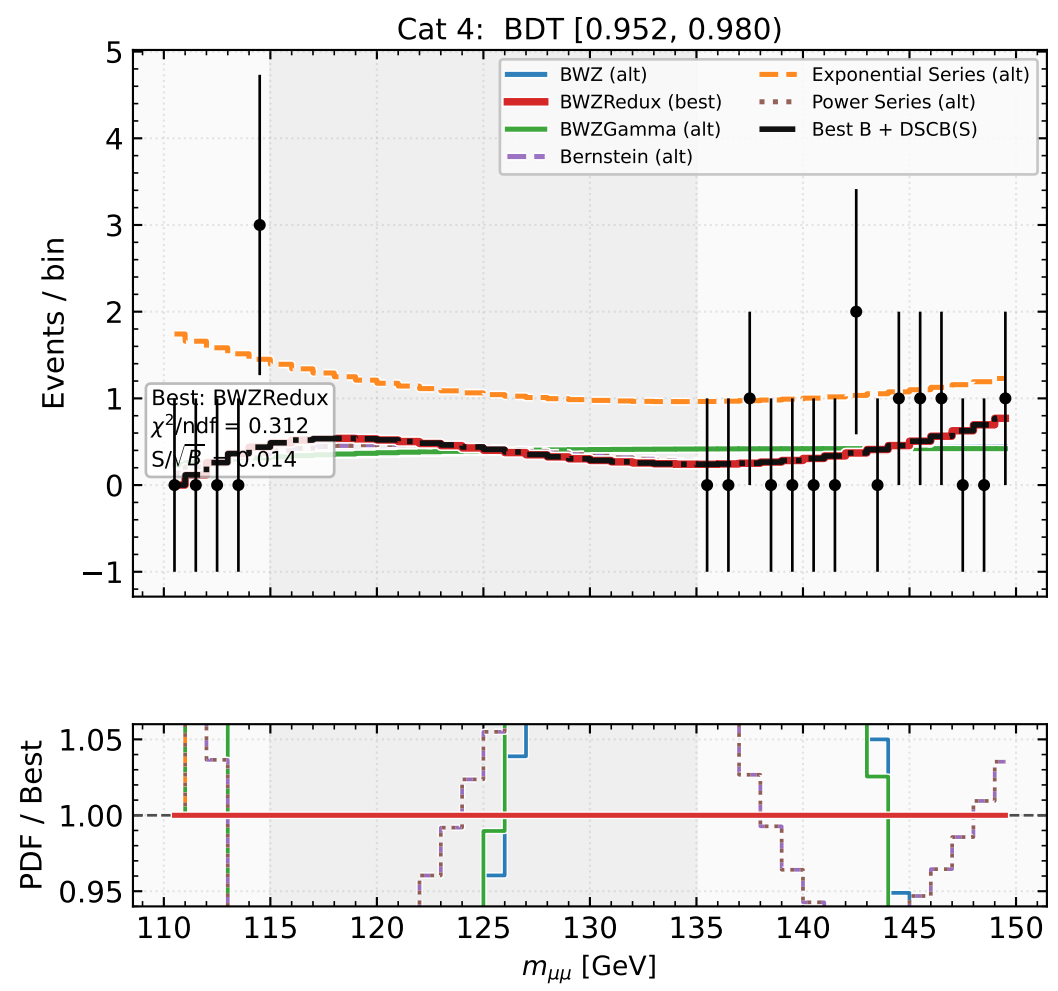
Per-category dimuon mass: background model comparison and ratio-to-best

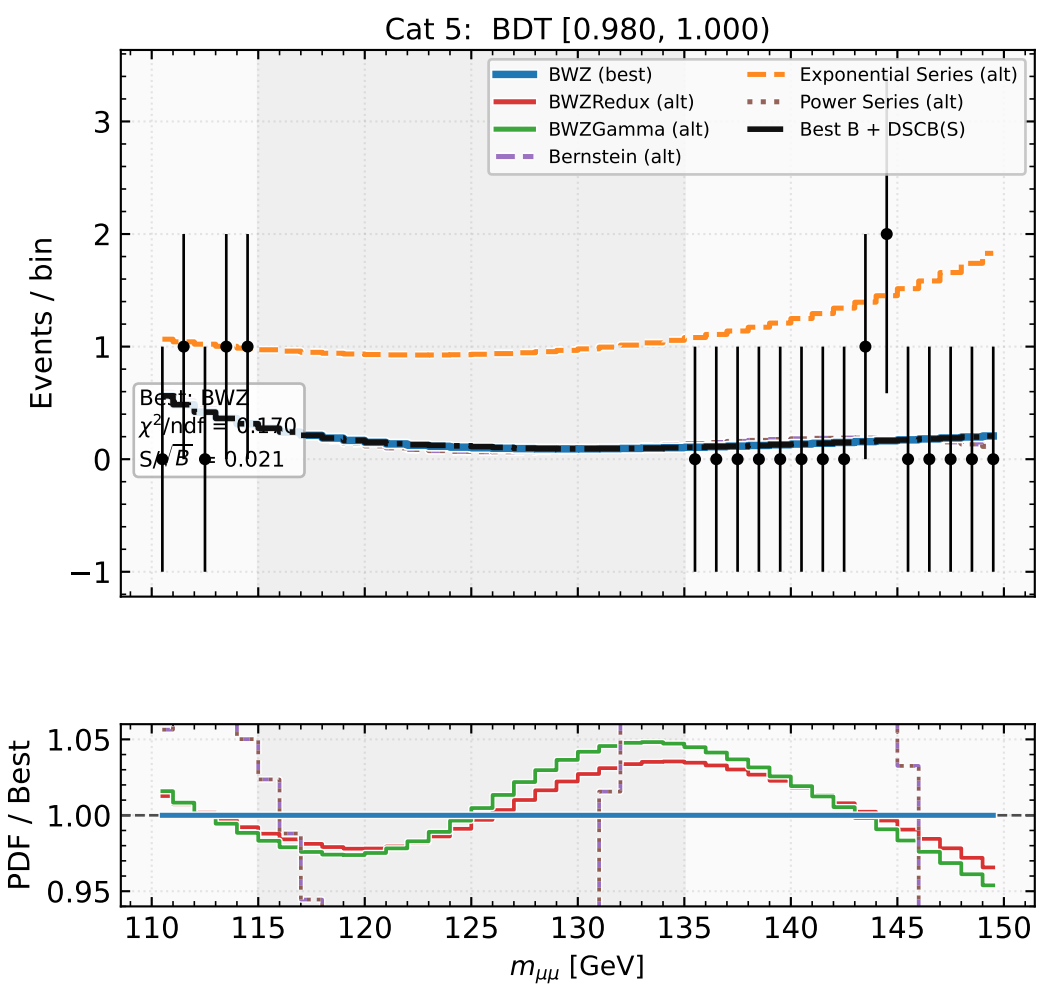












Per-category full-range fit comparison (chi2/ndf, ndf)

Category	BDT range	Best model	Best chi2/ndf	S/sqrt(B)	BWZ	BWZRedux	BWZGamma	Bernstein	Exponential Series	Power Series
0	[0.000, 0.531)	BWZ	0.786	0.002	0.786 (37)	0.806 (36)	0.787 (37)	0.845 (36)	0.825 (37)	0.845 (36)
1	[0.531, 0.787)	BWZ	0.922	0.004	0.922 (37)	0.948 (36)	0.923 (37)	0.949 (36)	0.993 (37)	0.949 (36)
2	[0.787, 0.898)	BWZGamma	0.895	0.007	0.899 (37)	0.908 (36)	0.895 (37)	0.945 (36)	1.270 (37)	0.945 (36)
3	[0.898, 0.952)	BWZRedux	0.529	0.008	0.530 (37)	0.529 (36)	0.531 (37)	0.541 (36)	0.854 (37)	0.541 (36)
4	[0.952, 0.980)	BWZRedux	0.312	0.014	0.327 (37)	0.312 (36)	0.327 (37)	0.321 (36)	0.975 (37)	0.321 (36)
5	[0.980, 1.000)	BWZ	0.170	0.021	0.170 (37)	0.175 (36)	0.170 (37)	0.172 (36)	1.203 (37)	0.172 (36)

Optimized category summary

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	Expected signif. (S/sqrt(B))
0	[0.000, 0.531)	585	643.31 ± 10.77	0.909	1364.06 ± 15.67	0.001
1	[0.531, 0.787)	87	68.69 ± 3.53	1.266	152.25 ± 5.29	0.004
2	[0.787, 0.898)	47	32.23 ± 2.44	1.458	59.01 ± 3.27	0.007
3	[0.898, 0.952)	29	13.18 ± 1.58	2.201	27.56 ± 2.30	0.010
4	[0.952, 0.980)	10	4.79 ± 0.87	2.089	13.93 ± 1.59	0.015
5	[0.980, 1.000)	6	1.80 ± 0.60	3.336	4.98 ± 0.96	0.025

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.531)	1225.77	0.80	135.34	2.15	1364.06	245.51
1	[0.531, 0.787)	136.75	0.79	14.31	0.40	152.25	27.97
2	[0.787, 0.898)	52.72	0.87	5.24	0.18	59.01	10.67
3	[0.898, 0.952)	24.59	0.72	2.16	0.09	27.56	5.28
4	[0.952, 0.980)	12.41	0.60	0.89	0.04	13.93	2.52
5	[0.980, 1.000)	4.24	0.51	0.23	0.01	4.98	0.93